

EPGSpeller: Lexicon-Free Silent Spelling Recognition with Electropalatography

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Abstract—Hands-free character entry motivates decoding beyond fixed command vocabularies; this paper reports an offline recognition-layer study on a corpus-bounded SilentSpeller spelling task, not interactive augmentative and alternative communication (AAC) deployment. We present EPGSpeller, which decodes character strings from SmartPalate-style electropalatographic contact time series using connectionist temporal classification and greedy decoding before lexicon projection. On four healthy participants, word-holdout (P1) and instance-holdout (P2) character error rate (CER) stay near 0.18, cross-user transfer (P3) exceeds 0.69, and one held-word example per label on p3 at k=1 partially recovers error to about 0.35 CER. Results are limited to offline recognition-layer evaluation; interactive AAC input is future work.

Index Terms—electropalatography, silent spelling, AAC, lexicon-free recognition, character-level decoding, calibration, silent speech interface

I. INTRODUCTION

For users and usage scenarios that cannot rely on speech or hand movement, hands-free character entry pathways are important. Silent speech interfaces are a promising direction for this requirement, and research has progressed on inferring linguistic intent from articulatory movements without vocalization, biosignals, and intraoral signals [1]–[3]. However, character entry for augmentative and alternative communication (AAC) requires more than recognizing a small set of predefined commands: users need to compose names, medication labels, idiosyncratic expressions, and short messages on the spot. This paper tests recognition-layer feasibility on a public healthy-participant release; interactive AAC deployment is out of scope (Section VI). Accordingly, bringing silent input closer to AAC requires spelling recognition—decoding character strings—rather than recognition confined to a closed vocabulary.

Existing silent speech interfaces have already demonstrated important capabilities. Studies report recognizing silent or low-volume speech intent using biosignals, perioral motion, tongue and lip imagery, in-ear sensing, and near-whisper input [4], [5]. These results show that silent signals can serve as practical input interfaces, but achieving high accuracy under constrained recognition is not the same as enabling users to spell unseen or context-dependent messages on the spot. AAC-oriented character entry requires not only high accuracy within a fixed vocabulary, but also character-level decoding that extends beyond that vocabulary.

Electropalatography (EPG) records tongue–palate contact during articulation as a multichannel time series [6]–[8] and is a natural signal for silent spelling (Section II, Table I). Each trial pairs a variable-length contact sequence with the word label the participant attempted to spell. SilentSpeller showed that SmartPalate-style EPG silent spelling can function as a portable, hands-free character entry modality [9]; we address the recognition layer that supports that modality—greedy string decoding before lexicon projection and the calibration bottleneck it entails. Silent command recognition, language-model-assisted character entry, and in-ear silent spelling with ReHEarSSE [5] are related, but the question here is how far spelling sequences can be decoded from EPG contact sequences without a fixed output lexicon.

Fixed-vocabulary constraints are especially important for AAC-oriented silent input. In fixed-vocabulary recognition, the words a system can output depend on a candidate set defined at training or evaluation time; dictionary constraints or language-model assistance can reduce errors but simultaneously restrict spellable words. Character-level silent spelling shifts the input problem from selecting a candidate word to composing a character string, so users are not limited to words that appeared as output classes during training. In this respect, character-level silent spelling recognition is an important pathway from fixed-command recognition toward an AAC-oriented character entry layer.

We present EPGSpeller, a recognizer that decodes character strings from EPG contact time series, to address this recognition problem. EPGSpeller is a sequence recognizer that maps contact frame sequences directly to character strings. To align variable-length contact sequences with string labels of differing length, we use connectionist temporal classification (CTC), which can learn variable-length inputs and string labels without frame-level character boundaries [10]. The primary evaluation uses greedy decoding without vocabulary lists, language models, or lexicon projection (Table III); lexicon-projected metrics are auxiliary.

The central claim is that, with per-user calibration, EPG silent spelling can support greedy string decoding on the SilentSpeller word inventory—word holdout within that set for P1, not open-vocabulary proper-name entry. When the same recognizer is transferred directly to a new user, performance degrades substantially, indicating that cross-user differences in contact patterns, articulatory style, and sensor placement strongly govern recognition performance. Accordingly, the central remaining challenge is neither the feasibility of the input modality nor inference speed, but how much calibration data

is sufficient per new user—calibration-efficient personalized recognition.

Experiments compare within-user decoding, an auxiliary SilentSpeller bridge diagnostic (not a direct era-to-era decoder comparison; Section IV), cross-user transfer, and low-shot calibration (Table V). Within-user protocols P1 and P2 test word-holdout and instance-holdout decoding when training data from the target user are available; P2 is an optimistic repetition upper bound. P3 tests zero-shot reuse on a new participant; multi-source transfer (P3MS) and preprocessing sweeps test whether simple normalization explains the cross-user gap. Low-shot calibration on p3 quantifies recovery when one or two held-word examples per label are available from the target user.

This paper makes three contributions. First, we formulate SmartPalate-style EPG silent spelling as a greedy string-decoding problem rather than fixed-command recognition, and present an evaluation protocol suite (P1, P2, P3, and k-shot) that compares within-user, unseen-word, cross-user transfer, and low-shot calibration conditions within a single task. Second, under this framework, we conduct a calibration feasibility study and quantify both the viability of the recognition layer under within-user conditions (P1, P2; Table VII) and the calibration bottleneck under low-shot target-user adaptation (k-shot; Table VIII). Third, we show that cross-user transfer (P3; Table VII) does not reach within-user performance even with multi-source transfer or preprocessing controls, and report that user dependence remains an essential calibration problem. The novelty of this paper lies not in the CTC recognizer architecture, but in this task and protocol formulation together with the calibration feasibility study.

II. SIGNAL, DATA, AND TASK DEFINITION

The recognizer receives not an acoustic waveform or video, but a multichannel time series representing tongue–palate contact, and outputs not a closed set of word classes but a character string.

A. EPG Contact Time Series

EPG records tongue–palate contact during articulation as a multichannel time series and has been used in speech production research, speech training, and articulatory analysis [6]–[8]. In the SmartPalate-style recordings considered here, each time-step frame represents a contact pattern indicating where on the palate contact occurs. Each trial is treated as a variable-length sequence of EPG contact frames, and the input received by EPSSpeller is this frame sequence.

In the experimental representation used in this paper, SmartPalate contact channels are mapped to a grid aligned with palate layout (a surrogate grid), and aligned contact channels are used. A surrogate grid assigns each channel to row–column coordinates on the palate, supporting both treating channels as independent scalar features and comparing front ends that exploit spatial layout on the palate. The physical SmartPalate device specification and the post-export experimental representation used in this paper are not identical.

TABLE I
SUMMARY OF SMARTPALATE-STYLE EPG SIGNALS
(EXPERIMENTAL REPRESENTATION).

Item	Value	Role
Input	Contact frames	Articulatory signal
Grid	16-by-16 surrogate gate	Spatial front ends
Channels	120 aligned	Multichannel patterns
Layout	4 missing, 5 dup.	Export mismatch
Output	Character string	Spelling, not classes
Decode	Greedy string	Lexicon-free eval.

Table I specifies the number of aligned channels, the surrogate grid, missing placements, and duplicate placements.

B. Samples and Labels

Each sample consists of a variable-length EPG contact sequence and the word label that the participant attempted to spell. Labels are treated as character strings, not as word classes. For example, when the target word of a trial is a particular English word, the model learns the character sequence rather than treating the word as a single class identifier. This design makes EPSSpeller a silent spelling recognizer rather than a finite word classifier.

In a word classifier, the output space is restricted to a vocabulary defined at training or evaluation time. In a string decoder, the recognizer can output character sequences and can therefore, in principle, compose words that never appeared as a single word class during training. The evaluation target is how accurately the character string corresponding to the target word can be decoded from the EPG contact time series.

For the primary evaluation, we use character-string output before lexicon projection; lexicon-projected metrics are auxiliary (Table III).

C. Dataset

This paper uses EPG recordings derived from SilentSpeller [9]. The data consist of silent spelling trials recorded with a SmartPalate-style intraoral sensor. The number of trials, word labels, sequence length, and mean contact rate differ across participants. These differences are important for interpreting later results. Within-user recognition measures performance when sufficient calibration data are available; cross-user transfer may manifest differences in contact rate, articulatory style, sensor placement, and oral anatomy as errors.

Table II summarizes the dataset for each participant. p1 and p2 each contain 2328 trials and 1164 word labels. p3 contains 2797 trials and 1164 word labels, with a lower mean contact rate than the other participants; p4 contains 1167 trials and 1162 word labels with lower repetition density. Protocol labels P1–P3 denote evaluation splits; participant ids p1–p4 denote SilentSpeller recordings (Table II). Contact rate (Cr.) is the

TABLE II

OVERVIEW OF THE SILENTSPELLER-DERIVED DATASET (PART.: PARTICIPANT; TRI.: TRIALS; LAB.: WORD LABELS; FR.: MEAN SEQUENCE LENGTH IN FRAMES; CR.: MEAN CONTACT RATE).

Part.	Tri.	Lab.	Fr.	Cr.
p1	2328	1164	234	0.19
p2	2328	1164	196	0.20
p3	2797	1164	283	0.12
p4	1167	1162	229	0.19

fraction of active palate channels per frame, averaged over trials; lower values indicate sparser contact patterns. These participant-level differences motivate treating protocols P1/P2 as within-user upper-bound conditions and P3 as the primary transfer bottleneck rather than pooling participants. Because of p4’s lower repetition density, the same calibration conditions as the other participants cannot be set for some within-user evaluations.

p1/p2 support within-user and calibration evaluation; p3 is the primary transfer and k-shot target; p4 is mainly a transfer target (lower repetition density).

For p1 and p2, repetitions per word are relatively balanced, making it easier to construct within-user unseen-word evaluation and low-shot calibration evaluation. By contrast, p4 has fewer trials relative to the number of word labels and cannot provide sufficient conditions with multiple repetitions of the same word. Therefore, p4 is treated mainly as a cross-user transfer target and is not mechanically included in all within-user and calibration conditions.

D. Task Definition and Metrics

The primary task in this paper is to decode a character string from an EPG contact time series. The input is a contact frame sequence of varying length. The output is a normalized alphabetic character string. The model is not given frame-level character boundaries and learns the correspondence between the contact sequence of an entire trial and the word label. Accordingly, the recognizer described in later sections uses the connectionist temporal classification (CTC) framework [10].

The primary metric is character error rate (CER) computed from greedy string decoding based on edit distance between the output string and the reference string before lexicon projection. Lexicon-projected error (LEX) and real-time factor (RTF) are supplementary diagnostics (Table III).

Evaluation conditions are divided into within-user calibration, unseen-word generalization, and cross-user transfer without target calibration; detailed definitions appear in Section IV (Table V).

III. EPGSPELLER RECOGNITION SYSTEM

This section defines the EPGSpeller recognition system. EPGSpeller provides a recognition layer for greedy character-string decoding on top of the input modality demonstrated by SilentSpeller [9].

TABLE III

TASK DEFINITION AND PRIMARY EVALUATION.

Item	Definition	Role
Input	EPG contact frames	Not acoustic
Target	Character string	Not word class
Primary decode	Greedy string	No lexicon
Primary metric	CER	Pre-lexicon
LEX	/ Projection / cost	Supplementary
RTF		

A. Overview and CTC Decoding

EPGSpeller is a sequence recognizer that takes electropalatographic contact frame sequences and outputs the corresponding character strings. Each trial is a variable-length contact sequence; character boundaries and phoneme boundaries are not provided. The primary output path is lexicon-free greedy character-string decoding defined in Section II-D.

In silent-spelling EPG sequences, the correspondence between each frame and a character is not explicit. Participants articulate continuously, and tongue–palate contact is not segmented at the character level. EPGSpeller uses connectionist temporal classification (CTC) for this alignment [10]. CTC learns sequence outputs without frame-level labels by collapsing frame-level outputs that include blank symbols and repetitions into the final character string.

B. Main Recognizer: Vector/GRU Path

In the main recognizer, each electropalatographic frame is treated as a flat contact vector. The temporal recognizer is a GRU-based recurrent CTC decoder. We treat this vector/GRU CTC path as the main recognizer throughout the paper. The central claim that electropalatographic silent spelling can support greedy character-string decoding when personalized calibration is available is evaluated on this main recognizer and greedy decoding.

Multichannel contact frames are mapped through an input projection into a representation that the sequence recognizer can process efficiently. Along the time axis, compression reduces sequence length while preserving local context. During training, we apply dropout, noise injection, smoothing, and

TABLE IV

BOUNDARY BETWEEN MAIN RECOGNIZER AND DESIGN ANALYSES.

Element	Claims?	Role
vector/GRU + greedy decode	Yes	Primary result
LEX projection	No	Closed-vocab. diag.
Spatial / electrode variants	No	Engineering analyses
Preprocessing controls	No	Transfer-gap control

SpecAugment-style temporal and channel regularization to improve robustness to contact-position variability, missing channels, and participant-specific differences [11].

C. Spatial Front Ends and Electrode Reduction

SmartPalate contact channels have a layout on the palate. We compare row-column, grid, and patch front ends to test whether spatial structure helps electropalatographic silent-spelling recognition. The row-column front end summarizes contact patterns on the surrogate palatal grid into row-wise and column-wise activity. Grid and patch front ends use local spatial structure more explicitly, but electropalatographic contact is sparse and strongly participant-dependent, so a strong two-dimensional spatial assumption is not always advantageous. Results in Section V show that the flat vector/GRU path remains competitive on P1.

We also compare contact-frequency-based selection, spatially dispersed selection, ranking transfer from other participants, and random subsets to examine whether recognition performance can be preserved with fewer channels. Electrode reduction and missing-channel robustness are engineering analyses; they are not primary means of solving cross-user transfer. Later analyses indicate that $K=64$ channels retain near-main P1/P2 performance for the selection methods tested.

D. Implementation Details

The main recognizer connects a CTC output layer to a bidirectional GRU (1024 units, 5 layers); input projection maps contact-channel representations to 256 dimensions. Along time, compression with kernel length 32 and stride 4 reduces sequence length. The output has 40 classes (26 alphabet letters, blank, CTC blank, and other special symbols). Training uses batch size 64, fixed learning rate 0.02, L2 regularization coefficient $1e-5$, dropout rate 0.4, and 10,000 training steps. Data augmentation applies SpecAugment-style temporal masking ($T=12$, rate 0.4), electrode masking ($F=6$, rate 0.4), and time warping ($W=0.05$, rate 0.2), plus noise and smoothing. Results are averaged over four trials with random seeds 0–3; mean and standard deviation are reported. In PCA variants, 120 contact channels are compressed to 16 principal components before input projection. Key hyperparameters are listed in the supplementary configuration table.

Spatial front ends, electrode reduction, and regularization are engineering design comparisons for representing and simplifying SmartPalate-style electropalatographic signals. Table IV separates the main recognizer from auxiliary analyses referenced later in the paper. The next section defines the evaluation conditions under which this recognizer is compared.

IV. EVALUATION SETUP AND DECODING CONDITIONS

This section fixes how EPGSpeller is compared across evaluation conditions and which metrics support the central claims. The core evaluation of EPGSpeller is character-string decoding without lexicon projection. Within-user recognition, unseen-word recognition, cross-user transfer, and low-shot calibration each answer a distinct question.

A. Evaluation Conditions

EPGSpeller evaluation separates recognition questions relevant to future AAC systems. Within-user conditions measure how accurately character strings can be decoded when the recognizer is trained on data from the target user. Word-holdout conditions measure whether spelling recognition generalizes to word labels not seen during training. Cross-user transfer conditions measure whether a recognizer trained on other users can be reused without target-user calibration data. Low-shot calibration conditions measure how much performance recovers when a few calibration examples are obtained from a new user.

P1 (word holdout, within user) is the primary evidence for lexicon-free spelling with personalized calibration. P2 (instance holdout, within user) is treated as a near-upper-bound within-user condition. P3 (cross-user transfer) is the primary evidence for the calibration bottleneck. Multi-source transfer, low-shot calibration, and the SilentSpeller bridge condition are auxiliary. Multi-source transfer (Multi-src; P3MS when multiple sources train before cross-user evaluation) trains on combinations of source participants before evaluating on a held-out target user. The SilentSpeller bridge (SS bridge) is a controlled unseen-word diagnostic on participant p1; it does not reproduce the original interactive SilentSpeller system. Direct comparison with the SilentSpeller-era recognizer and modern alternative decoders (beam search, language-model integration, attention-based encoder-decoder, and others) is outside the scope of this paper and is left for future work.

B. Metrics and Decoding Conditions

The primary metric is character error rate (CER) from greedy character-string decoding without lexicon projection. Lexicon-projected error (LEX) and real-time factor (RTF) are auxiliary diagnostics for closed-vocabulary conditions and inference cost; actual entry speed also depends on articulation rate, correction, feedback, and fatigue. Table V maps each condition to the practical question it answers; the rows are ordered from within-user evidence (P1/P2) through transfer (P3/P3MS) to calibration and the SilentSpeller bridge.

C. Low-Shot Calibration and Controls

In low-shot calibration evaluation, k -shot calibration obtains k examples per word label from the target user for words that appeared during training; it is distinct from generalization to unseen words not present during training (P1). Feasibility is constrained by per-word repetition density, not only by total trial count. We evaluate low-shot calibration within the range supported by the current data: $k=1$ and $k=2$ on p3; $k=4$ is limited to 14 words; $k=8$ is infeasible. These counts follow directly from per-word repetition in the public release and explain why higher- k curves cannot yet be reported as standard benchmarks.

The SilentSpeller bridge performs controlled unseen-word diagnosis under the same input modality (Section IV). Spatial front ends, electrode reduction, preprocessing controls, and lexicon projection support interpretation but do not replace P1–P3. P1 tests unseen-word generalization within a calibrated

user; P2 tests repetition recognition under optimistic within-user conditions; P3 tests transfer without target-user calibration. We do not report a pooled average across these conditions because each answers a different deployment question. Counts follow per-word repetition in the public SilentSpeller release (Table VI).

V. RESULTS

This section presents EPGSpeller results as a single narrative organized around the protocol map in Table V. We first compare within-user conditions with user calibration against cross-user transfer using the main vector/GRU recognizer and greedy string decoding before lexicon projection. We then connect to SilentSpeller-style unseen-word evaluation, evaluate low-shot calibration, multi-source transfer, and preprocessing controls, and summarize engineering analyses.

A. Main Results: Strong Within-User Performance, Large Cross-User Degradation

Table VII summarizes the main results. With the same vector/GRU recognizer and greedy string decoding, there is a large gap between within-user conditions with target-user calibration and cross-user transfer without it. P1 (word holdout, within user) reaches 0.1796 ± 0.0844 CER (mean $\pm$ std over four random seeds). P2 (instance holdout, within user) reaches 0.1451 ± 0.0627 CER (three seeds). In contrast, P3 (cross-user transfer) degrades to 0.6909 ± 0.1332 CER (six transfer directions). The P1/P2 mean $\pm$ std values are aggregated over random-seed repeats ($n=4$ and $n=3$, respectively); P3 is aggregated over transfer directions ($n=6$). Lexicon-projected values (LEX) are reported as a supplementary diagnostic for closed-vocabulary conditions. LEX further reduces error under within-user conditions, but the lexicon-free claim rests on CER before lexicon projection.

B. SilentSpeller Bridge and Low-Shot Calibration

We assess an auxiliary SilentSpeller bridge on participant p1 with 100 held-out test words from the same release.

TABLE V

EVALUATION CONDITION MAP. EACH CONDITION CORRESPONDS TO A DIFFERENT PRACTICAL QUESTION.

Cond.	Question	Split / role
P1	Can unseen words be spelled?	Word label; within-user
P2	Can another repetition be recognized?	Trial repetition; optimistic
P3	Can the model transfer as is?	Participant; bottleneck
P3MS	Do multiple sources help?	Multi-src transfer
Multi-src	Source combination label	Same as P3MS
k-shot	Does performance recover?	Cal. examples per word
SS	Unseen-word layer	100 held words on p1
bridge	strength?	

TABLE VI

LOW-SHOT CALIBRATION FEASIBILITY ON P3 (REQ.: REQUIRED REPETITIONS PER WORD; FEAS.: FEASIBLE VOCABULARY SIZE).

k	Req. rep./word	Feas. vocab	50-word eval
0	1	1164	Yes
1	2	1164	Yes
2	3	363	Yes
4	5	14	No
8	9	0	No

The vector/GRU recognizer reaches 0.1133 ± 0.0128 CER averaged over four seeds under this non-interactive setup. This sanity check confirms unseen-word decoding within the release inventory; it does not compare against SilentSpeller-era interactive decoders (deferred; Section IV).

Low-shot calibration experiments quantify cross-user transfer as a calibration-quantity problem. On p3, zero-shot single-source transfer remains at 0.8403 ± 0.0948 CER ($n=8$), and zero-shot multi-source transfer at 0.8381 ± 0.0336 CER ($n=4$). One example per word improves to 0.3503 ± 0.0600 CER ($n=4$); two examples per word to 0.2458 ± 0.0085 CER ($n=4$). Higher- k evaluation is constrained by public data density (Table VI).

C. Transfer Controls and Engineering Analyses

Transfer controls test whether multi-source training or simple preprocessing can close the cross-user gap without target-user calibration. Multi-source transfer partially improves cross-user performance but does not resolve the calibration problem. When p4 is the target user, single-source cross-user transfer with all training sources to p4 (P3) reaches 0.686 CER, whereas multi-source transfer (P3MS) reaches 0.582 CER—both remain far above the within-user P1/P2 range in Table VII (roughly 0.15–0.18 CER). Preprocessing controls test whether cross-user failure can be explained by simple scale differences or contact-rate mismatch alone. A sweep over raw input, standardization, contact-rate matching, target-neighborhood distribution normalization, and PCA-based standardization leaves best protocol averages at 0.6426 for P3 and 0.5692 for P3MS.

We analyze how SmartPalate signals should be represented and how future devices might be simplified. On P1, the vector/GRU path reaches 0.18 ± 0.08 CER; row-column pooling is close at 0.20 ± 0.09 , whereas grid and patch variants

TABLE VII

MAIN RESULTS FOR THE VECTOR/GRU RECOGNIZER (MEAN $\pm$ STD). N = SEEDS (P1/P2) OR DIRECTIONS (P3). LEX: LEXICON-PROJECTED CER; RTF: REAL-TIME FACTOR.

Cond.	CER	LEX	RTF
P1	0.1796 ± 0.0844	0.1018 ± 0.0679	0.000628 ± 0.000232
P2	0.1451 ± 0.0627	0.0747 ± 0.0476	0.000107 ± 0.000010
P3	0.6909 ± 0.1332	0.6442 ± 0.0604	0.000110 ± 0.000014

TABLE VIII

LOW-SHOT CALIBRATION AND EVALUABILITY ON P3 (CER MEAN $\pm$ STD).

Condition	CER	n
Zero-shot, single source	0.8403 ± 0.0948	8
Zero-shot, multi-source	0.8381 ± 0.0336	4
k=1	0.3503 ± 0.0600	4
k=2	0.2458 ± 0.0085	4
k=4	— (14 words)	—
k=8	infeasible	0

TABLE IX

ENGINEERING ANALYSES: SPATIAL FRONT ENDS AND ELECTRODE REDUCTION (GREEDY CER, MEAN $\pm$ STD).

(a) P1 spatial front end		(b) K=64 electrode reduction		
Front end	CER	Protocol	Method	CER
vector/GRU	0.18 ± 0.08	P1	top-k	0.195 ± 0.083
row-column	0.20 ± 0.09	P1	spatial	0.191 ± 0.078
grid	0.31 ± 0.15	P2	top-k	0.154 ± 0.059
patch	0.30 ± 0.07	P2	spatial	0.154 ± 0.062

degrade to roughly 0.31 ± 0.15 and 0.30 ± 0.07 (four seeds; Table IX(a)). Two-dimensional grid assumptions are therefore not consistently advantageous on sparse, user-specific contacts.

Electrode reduction at K=64 channels retains near-main P1/P2 performance across top-k, spatial, transfer-ranked, and random selection (Table IX(b)). Augmentation suppresses degradation from local missing contacts and channel dropout. Taken together, the spatial-front-end, electrode-reduction, and preprocessing sweeps bound representation and normalization choices but leave the P3 calibration gap essentially unchanged relative to within-user P1/P2 performance. None of these engineering analyses closes the P3 cross-user gap without target-user examples.

VI. DISCUSSION

A. Findings and Calibration Bottleneck

Cross-user failure likely reflects contact-rate mismatch, articulatory differences, and sensor placement rather than decoder architecture alone (Table IX). Multi-source training (P3MS) and preprocessing controls shift cross-user CER only marginally relative to the within-user gap in Table VII. Within-user CER near 0.15–0.18 indicates offline recognizer feasibility, not interactive usability; the binding bottleneck is calibration quantity, not inference speed (Tables VII and VIII).

B. Limitations and Threats to Validity

a) *Scope and interactive AAC.*: This paper reports an offline recognition-layer study on a corpus-bounded spelling task, not end-to-end AAC deployment (Section IV). Entry speed, correction interfaces, and user learning burden remain untested.

b) *Participant cohort.*: Four healthy participants limit generalization and clinical claims (Table II). Larger or clinical cohorts require new data collection beyond the public release.

c) *Offline evaluation protocol.*: Metrics use stored trials with greedy decoding, not live articulation with feedback. RTF in Table VII bounds inference cost only; interactive latency, fatigue, and correction cycles are out of scope.

d) *SilentSpeller bridge fidelity.*: The SS-bridge differs from the original interactive SilentSpeller setup (Section IV). It confirms unseen-word decoding on p1 within the release inventory but not that system’s interface or decoder stack.

e) *Cross-user calibration bottleneck.*: P3 without target-user calibration remains far above within-user P1/P2 despite P3MS and preprocessing sweeps (Tables VII and IX). User-independent deployment is unsupported; calibration quantity, not representation choice alone, binds performance.

f) *External decoder baselines.*: Comparisons to SilentSpeller-era recognizers, beam search, language-model integration, and attention encoder–decoders are deferred (Section IV). Spatial-front-end and electrode-reduction analyses bound in-task representation choices under the same CTC/greedy protocol instead.

g) *Low-shot and high-k evaluability.*: Per-word repetition density limits higher-k curves (Table VI); k=4 covers 14 words and k=8 is infeasible on p3. Reported k=1 and k=2 points still show large recovery from small target-user samples (Table VIII).

h) *Lexicon-free claim scope.*: LEX can be lower than greedy CER under closed-vocabulary conditions (Table VII). Primary claims use CER before lexicon projection; open-vocabulary spelling beyond the release inventory is not evaluated.

i) *Ethics and reproducibility.*: Evaluation follows the public release consent conditions; EPGSpeller is offline analysis only. An anonymized supplementary zip (`put_this_zip_to_your_agent_or_llm_chat.zip`; upload the entire archive to a coding agent or LLM session) maps headline results to splits and tabulated artifacts; the main text remains self-contained without it.

VII. CONCLUSION

We presented a staged P1–P3 evaluation protocol for lexicon-free EPG silent spelling that localizes feasibility to within-user calibration and identifies cross-user transfer without target-user examples as the binding bottleneck under the public SilentSpeller release.

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